

Introduction

Code Smells in Large Language Models for Code

Informed Consent Form

Research Procedures. The purpose of this research is to validate the usefulness of PSC (Propensity Smelly Score), a metric designed to measure the likelihood of Large Language Models (LLMs) at producing Code Smells. By gathering expert feedback from software engineering students, researchers, and industry practitioners, this study aims to assess whether PSC effectively captures the real-world risks of code smells in generated code. The findings will help bridge the gap between automated benchmarking metrics and expert judgment, ensuring that LLM-generated code is evaluated beyond traditional accuracy measures.

Risk. There are no foreseeable risks to participating in this research.

Benefits. There are no benefits to you as a participant other than further research in machine learning for software engineering.

Participation. You must be at least 18 years old to participate. Your participation is voluntary, and you may withdraw from the study at any time and for any reason. If you decide not to participate or if you withdraw from the study, there is no penalty or loss of benefits to which you are otherwise entitled. There are no costs to you or any other party. Your decision on whether or not to participate will not prejudice your future relations with The College of William & Mary.

Confidentiality. The data collected by this study will be confidential, including your responses. Any information obtained in connection with this study that can be identified with you will remain confidential and disclosed only with your permission. You will be assigned a code number to protect your identity, and all data will be kept secure. If you give us your permission by signing this document, we plan to disclose the results of the questionnaire in any publication resulting from this study. The disclosed results will not be personally identifiable (if needed, they will be anonymized). The de-identified data could be used for future research without additional consent from participants. The Institutional Review Board (IRB) and Protection of Human Subjects (PHSC) committees that monitor research on human subjects may inspect study records during internal auditing procedures and are required to keep all information confidential.

Contact. This research is being conducted by Dipin Khati (dkhati@wm.edu), Alejandro Velasco (svelascodimate@wm.edu), Daniel Rodriguez (dhrodriguezcar@wm.edu), and Dr. Denys Poshyvanyk (denys@cs.wm.edu). Questions about the study can be directed to

them. For questions about your rights as a participant, please contact Jennifer Stevens (jastev@wm.edu, (757)-221-3862).

Consent. *By signing you agree with the aforementioned terms and conditions of the study.*

Sign with your name below

Demographics

Section 1 - Demographic Information

1. How would you describe your primary role?

2. How many years of experience do you have in software development?

☐ 1–3 years

- ☐ 4–6 years
- ☐ 7–10 years
- ☐ More than 10 years

3. What is your level of expertise in python?

4. How familiar are you with the concept of code smells?

- ☐ Not familiar
- ☐ Somewhat familiar
- ☐ Familiar
- ☐ Very familiar
- ☐ Expert

5. Do you have a background in machine learning? Formal (college classes, degree, certification) or informal (self-learning or other training)?

6. How familiar are you with using Large Language Models (LLMs) like ChatGPT, Copilot, or others for code generation?

- ☐ Never used them
- ☐ Tried them once or twice
- ☐ Occasionally use them
- ☐ Frequently use them
- ☐ Use them regularly in my workflow

Explanation for CodeSmellEval

Measuring the presence of Code Smells with PSC

Please carefully review the following detail about our metric, PSC

Large Language Models (LLMs), such as ChatGPT, Copilot, and CodeLlama, have revolutionized software engineering tasks by automating code generation. While these models excel in generating syntactically correct and functional code, they often introduce *code smells*. Code smells are indicators of poor design or implementation choices that can negatively affect the code's readability, maintainability, and reliability.

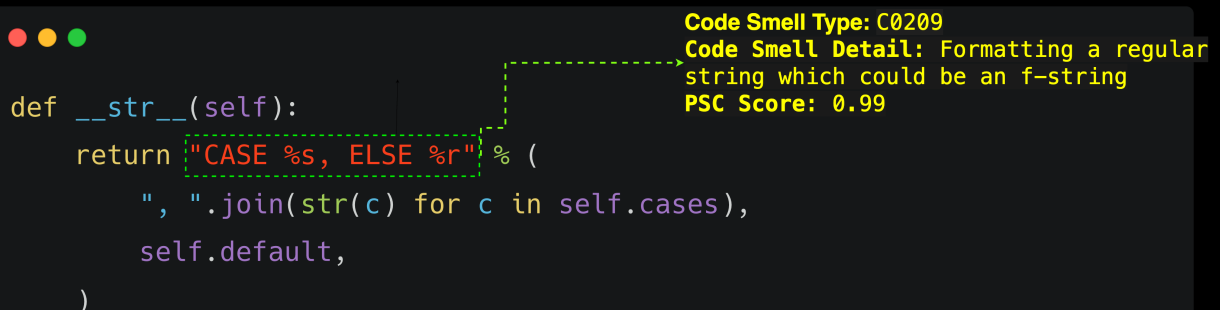
The **Propensity Smelly Score (PSC)** is a specialized metric that quantifies the likelihood of LLMs introducing code smells during code generation. It offers developers valuable insights into the reliability of LLMs and their tendency to produce specific types of code smells. A

higher PSC indicates a greater propensity to introduce a particular smell, while a lower score suggests a reduced likelihood.

Sample - S1_T

Sample 1

Please carefully review the following code snippet. We've highlighted a potential code smell in the generated code. Answer the questions below, using the provided information of the smell and its PSC value.



```
def __str__(self):  
    return "CASE %s, ELSE %r" % (  
        ", ".join(str(c) for c in self.cases),  
        self.default,  
    )
```

Code Smell Type: C0209
Code Smell Detail: Formatting a regular string which could be an f-string
PSC Score: 0.99

1. How likely do you think this code smell was introduced systematically or intentionally by the model (rather than being accidental)?

- ☐ Very unlikely
- ☐ Unlikely
- ☐ Neutral
- ☐ Likely
- ☐ Very likely

2. How important do you think it is to review or fix this code smell in a real-world codebase, considering the potential risk of similar issues reappearing in the future?

- ☐ Slightly important
- ☐ Moderately important
- ☐ Very important
- ☐ Extremely important

3. Did the PSC score influence your judgment about this code smell?

- ☐ Yes
- ☐ No
- ☐ Unsure

4. Please briefly explain how the PSC score influenced your decision?

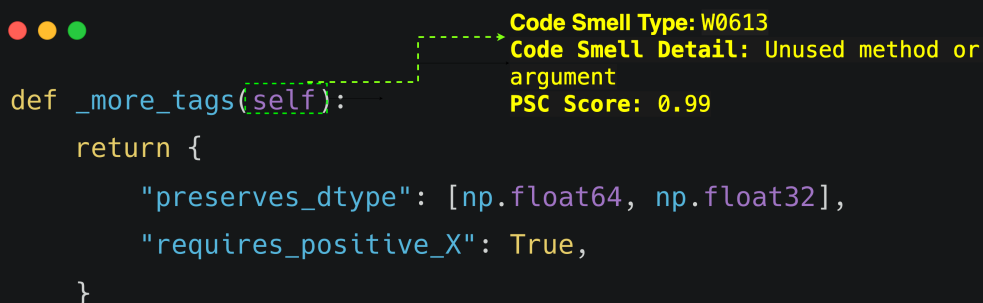
5. How confident are you in your judgment about this code smell?

- ☐ Not confident
- ☐ Slightly confident
- ☐ Moderately confident
- ☐ Very confident
- ☐ Completely confident

Sample - S2_T

Sample 2

Please carefully review the following code snippet. We've highlighted a potential code smell in the generated code. Answer the questions below, using the provided information of the smell and its PSC value.



The image shows a code editor with a dark background. A code snippet is displayed, and a dashed green line points from a text box to the 'self' argument in the function definition. The text box contains the following information:

- Code Smell Type: W0613
- Code Smell Detail: Unused method or argument
- PSC Score: 0.99

```
def _more_tags(self):  
    return {  
        "preserves_dtype": [np.float64, np.float32],  
        "requires_positive_X": True,  
    }
```

1. How likely do you think this code smell was introduced systematically or intentionally by the model (rather than being accidental)?

- ☐ Very unlikely
- ☐ Unlikely
- ☐ Neutral
- ☐ Likely
- ☐ Very likely

2. How important do you think it is to review or fix this code smell in a real-world codebase, considering the potential risk of similar issues reappearing in the future?

- ☐ Slightly important
- ☐ Moderately important
- ☐ Very important
- ☐ Extremely important

3. Did the PSC score influence your judgment about this code smell?

- ☐ Yes
- ☐ No

☐ Unsure

4. Please briefly explain how the PSC score influenced your decision?

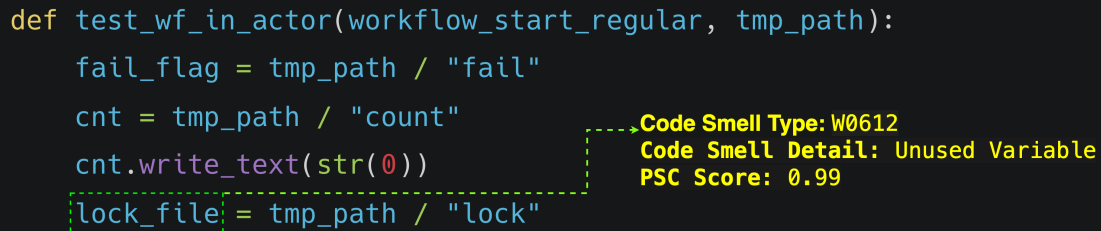
5. How confident are you in your judgment about this code smell?

- ☐ Not confident
- ☐ Slightly confident
- ☐ Moderately confident
- ☐ Very confident
- ☐ Completely confident

Sample - S3_T

Sample 3

Please carefully review the following code snippet. We've highlighted a potential code smell in the generated code. Answer the questions below, using the provided information of the smell and its PSC value.



```
def test_wf_in_actor(workflow_start_regular, tmp_path):  
    fail_flag = tmp_path / "fail"  
    cnt = tmp_path / "count"  
    cnt.write_text(str(0))  
    lock_file = tmp_path / "lock"
```

Code Smell Type: W0612
Code Smell Detail: Unused Variable
PSC Score: 0.99

1. How likely do you think this code smell was introduced systematically or intentionally by the model (rather than being accidental)?

- ☐ Very unlikely
- ☐ Unlikely
- ☐ Neutral
- ☐ Likely
- ☐ Very likely

2. How important do you think it is to review or fix this code smell in a real-world codebase, considering the potential risk of similar issues reappearing in the future?

- ☐ Slightly important
- ☐ Moderately important

- ☐ Very important
- ☐ Extremely important

3. Did the PSC score influence your judgment about this code smell?

- ☐ Yes
- ☐ No
- ☐ Unsure

4. Please briefly explain how the PSC score influenced your decision?

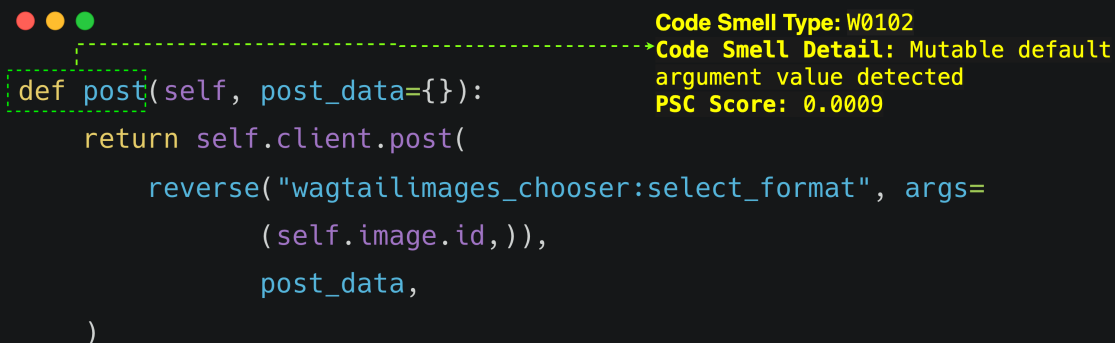
5. How confident are you in your judgment about this code smell?

- ☐ Not confident
- ☐ Slightly confident
- ☐ Moderately confident
- ☐ Very confident
- ☐ Completely confident

Sample - S4_T

Sample 4

Please carefully review the following code snippet. We've highlighted a potential code smell in the generated code. Answer the questions below, using the provided information of the smell and its PSC value.

A code editor window with a dark background. It shows a Python function definition:

```
def post(self, post_data={}):  
    return self.client.post(  
        reverse("wagtailimages_chooser:select_format", args=  
            (self.image.id,)),  
        post_data,  
    )
```

 The function signature `def post(self, post_data={}):` is enclosed in a dashed green box. A dashed green arrow points from this box to a yellow text box on the right. The yellow text box contains the following information:
Code Smell Type: W0102
Code Smell Detail: Mutable default argument value detected
PSC Score: 0.0009

1. How likely do you think this code smell was introduced systematically or intentionally by the model (rather than being accidental)?

- ☐ Very unlikely
- ☐ Unlikely
- ☐ Neutral
- ☐ Likely
- ☐ Very likely

2. How important do you think it is to review or fix this code smell in a

real-world codebase, considering the potential risk of similar issues reappearing in the future?

- ☐ Slightly important
- ☐ Moderately important
- ☐ Very important
- ☐ Extremely important

3. Did the PSC score influence your judgment about this code smell?

- ☐ Yes
- ☐ No
- ☐ Unsure

4. Please briefly explain how the PSC score influenced your decision?

5. How confident are you in your judgment about this code smell?

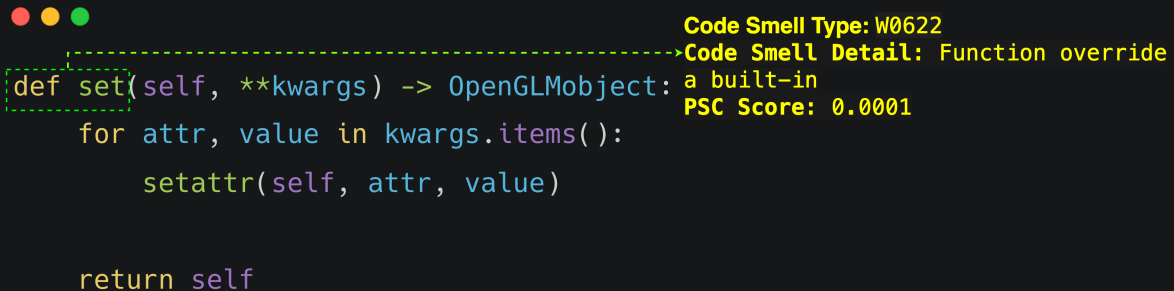
- ☐ Not confident
- ☐ Slightly confident
- ☐ Moderately confident
- ☐ Very confident

☐ Completely confident

Sample - S5_T

Sample 5

Please carefully review the following code snippet. We've highlighted a potential code smell in the generated code. Answer the questions below, using the provided information of the smell and its PSC value.



```
def set(self, **kwargs) -> OpenGLObject:
    for attr, value in kwargs.items():
        setattr(self, attr, value)

    return self
```

Code Smell Type: W0622
Code Smell Detail: Function override a built-in
PSC Score: 0.0001

1. How likely do you think this code smell was introduced systematically or intentionally by the model (rather than being accidental)?

☐ Very unlikely

- ☐ Unlikely
- ☐ Neutral
- ☐ Likely
- ☐ Very likely

2. How important do you think it is to review or fix this code smell in a real-world codebase, considering the potential risk of similar issues reappearing in the future?

- ☐ Slightly important
- ☐ Moderately important
- ☐ Very important
- ☐ Extremely important

3. Did the PSC score influence your judgment about this code smell?

- ☐ Yes
- ☐ No
- ☐ Unsure

4. Please briefly explain how the PSC score influenced your decision?

5. How confident are you in your judgment about this code smell?

- ☐ Not confident
- ☐ Slightly confident
- ☐ Moderately confident
- ☐ Very confident
- ☐ Completely confident

Overall Questions

Based on the previous 5 examples, please provide your opinion on following questions

1. In your opinion, does the presence of a code smell reduce your trust in LLM-generated code overall?

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Significantly
- ☐ Completely

2. If a tool for computing the PSC were available to assess potential issues or quality concerns in LLM-generated code, how likely would

you be to use it as part of your workflow (e.g., in your IDE or while using code generation tools)?

- ☐ Very unlikely
- ☐ Unlikely
- ☐ Neutral
- ☐ Likely
- ☐ Very likely

3. How should LLMs handle recurring code smells in code generations?

4. Would you prefer to see PSC scores for smells automatically flagged during LLM generation?

Yes
☐

No
☐

Not Sure
☐

5. What do you think is the biggest risk of using LLMs in software development?

Powered by Qualtrics